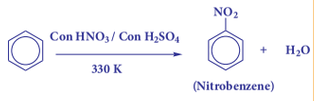
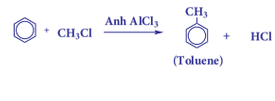
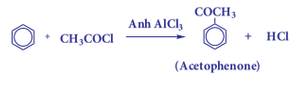
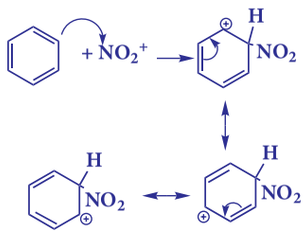
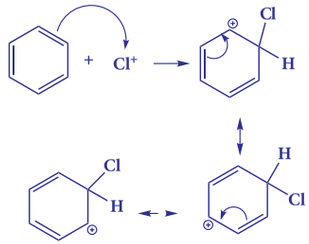
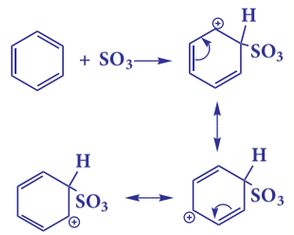
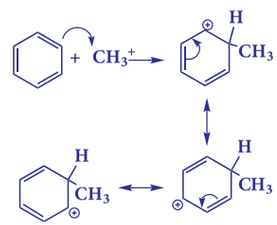
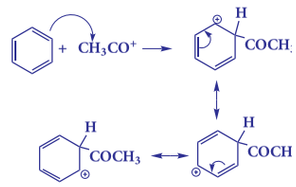
Types of reaction	NITRATION	HALOGENATION	SULPHONATION	FRIEDEL CRAFT'S ALKYLATION	FRIEDEL CRAFT'S ACYLATION
Reagents	Con HNO_3 + Con H_2SO_4	X_2 / Al X_3	Fuming H_2SO_4	CH_3Cl + anhydrous $AlCl_3$	CH_3COCl + anhydrous $AlCl_3$
Electro-phile	$-NO_2^+$	$-X^+$ (X = Cl, Br)	$-SO_3$ (Neutral electrophile)	$-\overset{\oplus}{C}H_3$	$-\overset{\oplus}{C}OCH_3$
Over all reaction					
Mechanism Step 1	$HNO_3 + H_2SO_4 \rightarrow \overset{\oplus}{N}O_2 + H\overset{\ominus}{S}O_4 + H_2O$	$AlCl_3 + Cl_2 \rightarrow \overset{\oplus}{Cl} + \overset{\ominus}{A}lCl_4$	$2H_2SO_4 \rightarrow H_3\overset{\oplus}{O} + SO_3 + H\overset{\ominus}{S}O_4$	$AlCl_3 + CH_3Cl \rightarrow \overset{\oplus}{C}H_3 + \overset{\ominus}{A}lCl_4$	$AlCl_3 + CH_3COCl \rightarrow CH_3\overset{\oplus}{C}O + \overset{\ominus}{A}lCl_4$
Step 2					
Step 3	